# ZHI YANG

+1-757-503-6096 | zyang23@wm.edu | ruayz.github.io

Research interests: Trustworthy AI, Multimodal Learning, Interactive AI Systems.

## EDUCATION

- **William & Mary (W&M)** *Fall 2026 - Present*
  *Ph.D. in Applied Science (Data Science concentration), first-year student* Williamsburg, VA, USA
  ◦ Supervisor: Jindong Wang

- **Southern University of Science and Technology (SUSTech)** *Sep. 2023 - Jun. 2026*
  *Master's degree in Computer Science and Technology* Shenzhen, China
  ◦ Supervisor: Changwu Huang, Ke Tang, Xin Yao.

- **Yunnan University (YNU)** *Sep. 2019 - Jun. 2023*
  *Bachelor's degree in Software Engineering* Kunming, China

## PUBLICATIONS

C=CONFERENCE, S=IN SUBMISSION

**[S.1]** Zhi Yang, Changwu Huang, Ke Tang, Xin Yao. **On the Fairness of Privacy Protection: Measuring and Mitigating the Disparity of Group Privacy Risk for Differentially Private Machine Learning**. arXiv preprint arXiv:2510.09114.

**[C.1]** Zhi Yang, Changwu Huang, Xin Yao. **Towards Private and Fair Machine Learning: Group-Specific Differentially Private Stochastic Gradient Descent with Threshold Optimization**. *The International Conference on Neural Information Processing (ICONIP)*. Singapore: Springer Nature Singapore, 2024: 66-80.

**[C.2]** Zhi Yang, Ziming Wang, Changwu Huang, Xin Yao. **An Explainable Feature Selection Approach for Fair Machine Learning**. *International Conference on Artificial Neural Networks (ICANN)*. Cham: Springer Nature Switzerland, 2023: 75-86.

## RESEARCH EXPERIENCE

- **Understanding and Evaluating Streaming Multimodal Models** *Sep. 2026 - Present*
  *Ongoing Research*
  ◦ Investigating how streaming adaptation affects the capabilities and reliability of vision-language models. This project systematically **compares streaming models with their offline counterparts** across image and video understanding tasks, examining changes in visual perception, temporal reasoning, robustness, safety, and fairness. We further study how streaming post-training, memory mechanisms, and interaction patterns contribute to these changes.

- **Impact of Visual Redundancy on Multimodal Reasoning** *Jan. 2025 - May. 2026*
  *Concluded Project*
  ◦ Investigated the impact of visual redundancy on multimodal reasoning through an exhaustive patch-level ablation study, partitioning images into $3 \times 3$ grids and evaluating all 512 possible patch combinations. Observed that a single patch was often sufficient for correct predictions, while additional visual information could lead to incorrect answers, revealing a **non-monotonic relationship between visual context and VLM performance**.

- **Adaptive LLM Personalization via Activation Engineering** *Oct. 2025 - Feb. 2026*
  *Concluded Project*
  ◦ Explored **rapid LLM personalization through activation engineering**, aiming to mitigate the cold-start problem without resource-intensive fine-tuning; Investigated the composition of steering vectors for fine-grained adaptation to diverse user backgrounds, revealing challenges in achieving precise and consistent control over response styles.

- **Emotion Influence on LLM-Based Multi-Agent Systems** *Jun. 2025 - Sep. 2025*
  *Concluded Project[]*
  ◦ Investigated how emotional stimuli influence agent behavior in multi-agent LLMs systems across diverse network topologies, finding no consistent sensitivity patterns across node positions or propagation structures, and showing that **modern LLMs remain robust against emotion-induced perturbations**.

- **Fair Privacy [S.1]** *Nov. 2024 - May. 2025*
  *Master's Thesis []*
  ◦ Summary: We address the question of whether AI systems expose different subgroups to unequal privacy risks by proposing **a stricter measurement for group privacy risk and an effective mitigation strategy**. Existing methods are either inaccurate or computationally expensive. We introduce a novel membership inference game to efficiently estimate worst-case privacy risks, enabling a more reliable assessment of disparities across groups; To enhance fairness, we develop a group-specific adaptive gradient clipping method for DP-SGD that reduces differences in privacy risks among subgroups.

- **Fairness & Privacy [C.1]** *Mar. 2024 - Jul. 2024*
  *Master's Thesis [ ]*
  - Summary: Recent research highlights the challenges of integrating differential privacy (DP) with group fairness. Existing methods typically address either the amplified accuracy disparities across sensitive groups caused by DP or the potential outcome unfairness in DP-trained models, often in isolation. We propose a group-specific DP-SGD training framework combined with classification threshold optimization, which **jointly reduces accuracy disparities while achieving outcome fairness**.
- **Explainability & Fairness [C.2]** *Feb. 2023 - Jun. 2023*
  *Bachelor's Thesis*
  - Summary: In this work, we propose an **explainable feature selection (ExFS) method** to enhance model outcome fairness by recursively removing features that contribute to unfairness, guided by the **feature attribution explanations (FAE) of the model's predictions**.

## Honors and Awards

- Outstanding Academic Scholarship *Oct. 2025*
- Outstanding Research Assistance Scholarship *Oct. 2025*
- **Outstanding Graduate Teaching Assistant** *Sep. 2024*
- Outstanding Research Assistance Scholarship *Sep. 2024*
- **National Scholarship (Top 1% in School of Software)** *Dec. 2022*
- National Silver Award, 8th Internet+ Innovation and Entrepreneurship Competition *Nov. 2022*
- University-level First-class Scholarship *Dec. 2021*
- University-level First-class Scholarship *Dec. 2020*